

A Project on
**Credit Card Fraud Detection Using
R Language**

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*A report submitted in partial fulfillment
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INTRODUCTION



In today's modern era, credit cards have become an essential part of our everyday lives in the digital age. They offer unmatched convenience for various transactions, such as online shopping and in store purchases. However, with the widespread use of credit cards comes a significant challenge; credit card fraud. This type of fraud poses serious threats to both financial institutions and cardholders, resulting in substantial financial losses and the potential compromise of personal information.

To address this ever-present challenge, machine learning has emerged as a valuable tool in the fight against credit card fraud. Backed by advanced data analytics, machine learning techniques have revolutionized how financial institutions and payment processors detect and prevent fraudulent activities. By analyzing vast amounts of transaction data, machine learning models can identify patterns and anomalies that indicate potential fraudulent behavior, ultimately enhancing the security of credit card transactions.

AIM OF THE PROJECT

- Predicting a transaction to be fraud with logistic regression, decision tree models.
- To compare the accuracy level of all the models to understand the best one.

DATASET

The dataset taken from Kaggle (link: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>), naming "creditcard.csv " comprises credit card transactions conducted by European cardholders in September 2013. This dataset covers transactions over two days, revealing a total of 492 frauds among 284,807 transactions. These transactions are classified as either fraudulent (marked with '1') or genuine (marked with '0') in the 'class' column. The dataset exhibits a severe class imbalance, with frauds accounting for only 0.172% of all transactions. It exclusively contains numerical input variables, a result of applying Principal Component Analysis (PCA) transformation. Regrettably, due to confidentiality constraints, the original features and additional context regarding the data remain undisclosed. The features 'V1' - 'V28' correspond to principal components derived from PCA, while 'Time' and 'Amount' are the only attributes that haven't undergone PCA transformation. Following are the information about the rest of the variables:

Sl. No.	Variable Names	Data Type	Description
1.	Time	float64	Feature that represents the time elapsed, in seconds, between each transaction and the initial transaction recorded in the dataset.
2.	Amount	float64	Transaction amounts.
3.	Class	int64	Classifies transaction as fraudulent or genuine.

First few rows of the dataset:

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22	V23	V24	V25	V26	V27	V28	Amount	class
0	0	-1.3598	-0.0728	2.53635	1.37816	-0.3383	0.46239	0.2396	0.0987	0.36379	...	-0.0183	0.27784	-0.1105	0.06693	0.12854	-0.1891	0.13356	-0.0211	149.62	0
1	0	1.19186	0.26615	0.16648	0.44815	0.06002	-0.0824	-0.0788	0.0851	-0.2554	...	-0.2258	-0.6387	0.10129	-0.3398	0.16717	0.1259	-0.009	0.01472	2.69	0
2	1	-1.3584	-1.3402	1.77321	0.37978	-0.5032	1.8005	0.79146	0.24768	-1.5147	...	0.248	0.77168	0.90941	-0.6893	-0.3276	-0.1391	-0.0554	-0.0598	378.66	0
3	1	-0.9663	-0.1852	1.79299	-0.8633	-0.0103	1.2472	0.23761	0.37744	-1.387	...	-0.1083	0.00527	-0.1903	-1.1756	0.64738	-0.2219	0.06272	0.06146	123.5	0
4	2	-1.1582	0.87774	1.54872	0.40303	-0.4072	0.09592	0.59294	-0.2705	0.81774	...	-0.0094	0.79828	-0.1375	0.14127	-0.206	0.50229	0.21942	0.21515	69.99	0

Summary of the Data

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9
count	284807	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05
mean	94813.8596	1.76E-12	-8.25E-13	-9.65E-13	8.32E-13	1.65E-13	4.25E-13	-3.05E-13	8.78E-14	-1.18E-12
std	47488.146	1.96E+00	1.65E+00	1.52E+00	1.42E+00	1.38E+00	1.33E+00	1.24E+00	1.19E+00	1.10E+00
min	0	-5.64E+01	-7.27E+01	-4.83E+01	-5.68E+00	-1.14E+02	-2.62E+01	-4.36E+01	-7.32E+01	-1.34E+01
25%	54201.5	-9.20E-01	-5.99E-01	-8.90E-01	-8.49E-01	-6.92E-01	-7.68E-01	-5.54E-01	-2.09E-01	-6.43E-01
50%	84692	1.81E-02	6.55E-02	1.80E-01	-1.98E-02	-5.43E-02	-2.74E-01	4.01E-02	2.24E-02	-5.14E-02
75%	139320.5	1.32E+00	8.04E-01	1.03E+00	7.43E-01	6.12E-01	3.99E-01	5.70E-01	3.27E-01	5.97E-01
max	172792	2.45E+00	2.21E+01	9.38E+00	1.69E+01	3.48E+01	7.33E+01	1.21E+02	2.00E+01	1.56E+01

	V21	V22	V23	V24	V25	V26	V27	V28	Amount	class
count	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	2.85E+05	284807	284807
mean	-3.41E-13	-5.72E-13	-9.73E-13	1.46E-12	-6.99E-13	-5.62E-13	3.33E-12	-3.52E-12	88.34962	0.00173
std	7.35E-01	7.26E-01	6.24E-01	6.06E-01	5.21E-01	4.82E-01	4.04E-01	3.30E-01	250.1201	0.04153
min	-3.48E+01	-1.09E+01	-4.48E+01	-2.84E+00	-1.03E+01	-2.60E+00	-2.26E+01	-1.54E+01	0	0
25%	-2.28E-01	-5.42E-01	-1.62E-01	-3.55E-01	-3.17E-01	-3.27E-01	-7.08E-02	-5.30E-02	5.6	0
50%	-2.95E-02	6.78E-03	-1.12E-02	4.10E-02	1.66E-02	-5.21E-02	1.34E-03	1.12E-02	22	0
75%	1.86E-01	5.29E-01	1.48E-01	4.40E-01	3.51E-01	2.41E-01	9.10E-02	7.83E-02	77.165	0
max	2.72E+01	1.05E+01	2.25E+01	4.58E+00	7.52E+00	3.52E+00	3.16E+01	3.38E+01	25691.16	1

INITIAL INSIGHTS OF THE DATA

dim() function in R Language is used to get or set the dimension of the data and str() is a diagnostic function that displays the internal structure of an R object.

```
> dim(df)
[1] 284807 31
```

```
> str(df)
'data.frame': 284807 obs. of 31 variables:
 $ Time : num 0 0 1 1 2 2 4 7 9 ...
 $ V1 : num -1.36 1.192 -1.358 -0.966 -1.158 ...
 $ V2 : num -0.0728 0.2662 -1.3402 -0.1852 0.8777 ...
 $ V3 : num 2.536 0.166 1.773 1.793 1.549 ...
 $ V4 : num 1.378 0.448 0.38 -0.863 0.403 ...
 $ V5 : num -0.3383 0.06 -0.5032 -0.0103 -0.4072 ...
 $ V6 : num 0.4624 -0.0824 1.8005 1.2472 0.0959 ...
 $ V7 : num 0.2396 -0.0788 0.7915 0.2376 0.5929 ...
 $ V8 : num 0.0987 0.0851 0.2477 0.3774 -0.2705 ...
 $ V9 : num 0.364 -0.255 -1.515 -1.387 0.818 ...
 $ V10 : num 0.0908 -0.167 0.2076 -0.055 0.7531 ...
 $ V11 : num -0.552 1.613 0.625 -0.226 -0.823 ...
 $ V12 : num -0.6178 1.0652 0.0661 0.1782 0.5382 ...
 $ V13 : num -0.991 0.489 0.717 0.508 1.346 ...
 $ V14 : num -0.311 -0.144 -0.166 -0.288 -1.12 ...
 $ V15 : num 1.468 0.636 2.346 -0.631 0.175 ...
 $ V16 : num -0.47 0.464 -2.89 -1.06 -0.451 ...
 $ V17 : num 0.208 -0.115 1.11 -0.684 -0.237 ...
 $ V18 : num 0.0258 -0.1834 -0.1214 1.9658 -0.0382 ...
 $ V19 : num 0.404 -0.146 -2.262 -1.233 0.803 ...
 $ V20 : num 0.2514 -0.0691 0.525 -0.208 0.4085 ...
 $ V21 : num -0.01831 -0.22578 0.248 -0.1083 -0.00943 ...
 $ V22 : num 0.27784 -0.63867 0.77168 0.00527 0.79828 ...
 $ V23 : num -0.11 0.101 0.909 -0.19 -0.137 ...
 $ V24 : num 0.0669 -0.3398 -0.6893 -1.1756 0.1413 ...
 $ V25 : num 0.129 0.167 -0.328 0.647 -0.206 ...
 $ V26 : num -0.189 0.126 -0.139 -0.222 0.502 ...
 $ V27 : num 0.13356 -0.00898 -0.05535 0.06272 0.21942 ...
 $ V28 : num -0.0211 0.0147 -0.0598 0.0615 0.2152 ...
 $ Amount: num 149.62 2.69 378.66 123.5 69.99 ...
 $ class : int 0 0 0 0 0 0 0 0 0 0 ...
```

We use table() function to check class imbalance and prop.table() function to compute percentages of the proportions

```
8 # checking class imbalance
9 table(df$class)
10 # class imbalance in percentage
11 prop.table(table(df$class))
12
13 labels <- c("legit", "fraud")
```

Output:

```
> # checking class imbalance
> table(df$class)

 0      1
284315  492
> # class imbalance in percentage
> prop.table(table(df$class))

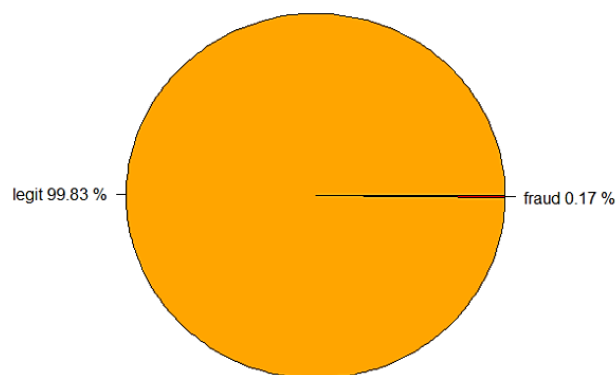
      0      1
0.998272514 0.001727486
```

DATA VISUALIZATION

This visualization is particularly valuable for conveying the class distribution within the dataset, aiding in the identification of patterns and irregularities. The subsequent insights derived from this visualization can contribute to a better understanding of the dataset and inform decision-making processes, especially in the context of fraud detection and credit card transaction analysis.

```
12
13 labels <- c("legit","fraud")
14 labels <- paste(labels, round(100*prop.table(table(df$class)),2))
15 labels <- paste(labels, "%")
16 pie(table(df$class),labels, col = c("orange", "red"),
17     main = "Pie chart of Credit Card Transactions")
18
```

Pie chart of Credit Card Transactions



The pie chart serves as a concise and intuitive representation, offering a quick overview of the relative proportions of fraud and legitimate transactions.

HANDLING MISSING VALUES

The dataset was checked for missing values but none were found in any of the columns.

```
6 # checking missing values
7 sum(is.null(df))
```

Output:

```
> # checking missing values
> sum(is.null(df))
[1] 0
```

TARGET VARIABLE DISTRIBUTION

The data is split with respect to two classes fraud and legit. Since 1 represent fraud, the variable class containing 1 is filtered into variable fraud and similarly class variable containing value 0 is filtered into variable legit.

```
18
19 #Change 'class' variable to factor
20 df$class <- as.factor(df$class)
21 levels(df$class) <- c("Legit", "Fraud")
22
23 # class ratio initially
24 table(df$class)
25
```

Output:

```
> #Change 'class' variable to factor
> df$class <- as.factor(df$class)
> levels(df$class) <- c("Legit", "Fraud")
> # class ratio initially
> table(df$class)

  Legit  Fraud
284315   492
> |
```

Here we can see that out of 284807 transaction 284315 are legit transaction and 492 are fraudulent.

DATA SPLITTING

Data splitting stands as a foundational procedure in both machine learning and data analysis. It revolves around the division of a dataset into multiple subsets, each serving distinct purposes. In this context, the primary division occurs between a training set and a testing set. The training set functions as the platform for training a machine learning model, while the testing set is set aside specifically for the purpose of assessing the model's performance. This separation is paramount in assuring the model's capacity to generalize effectively when presented with new, unseen data.

```
26 set.seed(123)
27 split <- sample.split(df$class, splitRatio = 0.7)
28 train <- subset(df, split == TRUE)
29 test <- subset(df, split == FALSE)
30
```

Here, `sample.split` is used to generate a logical vector (`split`) indicating which observations should be in the training set (`TRUE`) and which should be in the testing set (`FALSE`). The `splitRatio` parameter to set the desired ratio for the split. We split the dataset in 0.7:0.2 ratio for training data and testing data respectively, meaning 70% of data is used for training and rest for testing. Then, the `subset` function is used to create the actual training and testing sets based on this split.

BALANCING DATASET

Balancing a dataset is a crucial step in data preprocessing, especially in the context of machine learning, where it plays a pivotal role in addressing the challenges posed by imbalanced datasets. This process involves adjusting the class distribution, primarily achieved through oversampling the minority class, undersampling the majority class, or utilizing a combination of both techniques. Its significance lies in ensuring that the machine learning model does not exhibit bias toward the majority class, thereby enabling it to effectively learn from both classes. Consequently, this contributes to the model's ability to make precise predictions while reducing the risk of misclassification. This becomes particularly pertinent in scenarios where one class is rare or of specific interest.

Upon examining the dataset in a previous step of our analysis, it becomes evident that the data exhibits a pronounced class imbalance. Out of a total of 284,807 transactions, a staggering 284,315 are legit transactions, while only a mere 492 are flagged as fraudulent. We observe that the training data of the dataset is much more imbalanced. Hence, to rectify this imbalance, we chose to perform downsampling. This approach involves maintaining the entirety of the data in the minority class while reducing the size of the majority class. While undersampling has its drawbacks, including potential information loss, it remains a widely adopted and crucial step in data preprocessing. This method significantly contributes to enhancing the accuracy and reliability of machine learning models when working with imbalanced datasets.

```
30
31 # downsampling
32 set.seed(9560)
33 down_train <- downsample(x = train[, -ncol(train)],
34                           y = train$class)
35 table(down_train$class)
36
```

Output:

```
> table(down_train$class)

Legit Fraud
344     344
```

LOGISTIC REGRESSION

Logistic regression, a supervised machine learning technique, finds its primary application in tackling classification tasks. This statistical algorithm delves into examining the connection between a group of independent variables and the dependent binary variables. It stands as a robust tool for aiding decision-making processes, exemplified by its effectiveness in scenarios such as distinguishing between email spam and non-spam messages.

Logistic regression uses the sigmoid function to transform a linear combination of the input features into a value between 0 and 1. The sigmoid function has an S-shaped curve and maps any real-valued number to the range [0, 1]

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

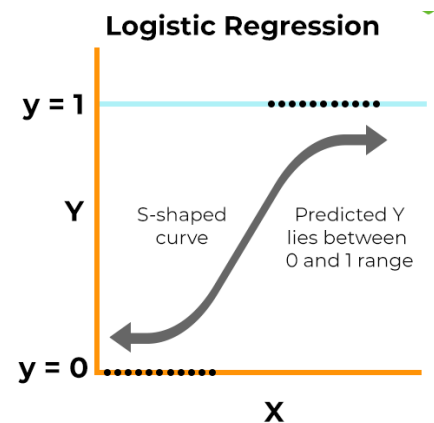
Where z is the linear combination of input features and their corresponding weights, plus a bias term:

$$Z = w_0 + w_1 * x_1 + w_2 * x_2 + \dots + w_r * x_r$$

Where:

w_0 is the bias term (intercept),

$w_1, w_2, \dots, w_r$ are the coefficients (weights) for each input feature $x_1, x_2, \dots, x_r$, and r is the number of input features.

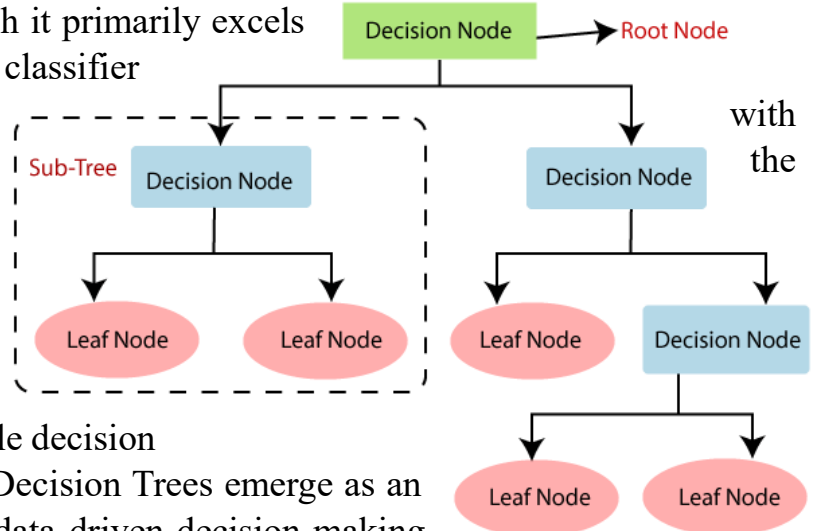


```
54
55 #logistic
56 glm_fit <- glm(Class ~ ., data = down_train, family = 'binomial')
57
58 pred_glm <- predict(glm_fit, newdata = test, type = 'response')
59
60 roc(test$class, pred_glm, plotit = FALSE)
61
62
```

DECISION TREE CLASSIFIER

The Decision Tree stands as a supervised learning approach, versatile enough to address both classification and regression tasks, although it primarily excels in handling classification challenges. This classifier

takes on the form of a tree-like structure, internal nodes representing features within dataset, branches embodying decision rules, and each leaf node offering a prediction for the outcome. The overarching goal is to build a model that can accurately forecast the value of the target variable by extracting clear and simple decision rules from the data's features. In essence, Decision Trees emerge as an interpretable and potent methodology for data-driven decision-making across a diverse spectrum of applications.



```
36
37 #CART Model Performance on imbalanced data
38 set.seed(5627)
39
40 orig_fit <- rpart(class ~ ., data = train)
41
42 #Evaluate model performance on test set
43 pred_orig <- predict(orig_fit, newdata = test, method = "class")
44
45 roc(test$class, pred_orig[,2], plotit = FALSE)
46 pred_down <- predict(down_fit, newdata = test)
47
48 set.seed(5627)
49 # Build down-sampled model
50 down_fit <- rpart(class ~ ., data = down_train)
51 pred_down <- predict(down_fit, newdata = test)
52 print('Fitting model to downsampled data')
53 roc(test$class, pred_down[,2], plotit = FALSE)
54
```

MODEL EVALUATION

Accuracy is a metric used to evaluate the performance of a classification model. It represents the ratio of correctly predicted instances to the total instances in a dataset. In other words, accuracy measures how well a model correctly classifies both positive and negative instances.

Accuracy serves as a fundamental metric, quantifying the overall correctness of our model's predictions. Specifically, it measures the ratio of correctly classified instances (both fraud and legitimate transactions) to the total number of transactions. A high accuracy percentage indicates a robust ability of the model to make correct predictions across both classes.

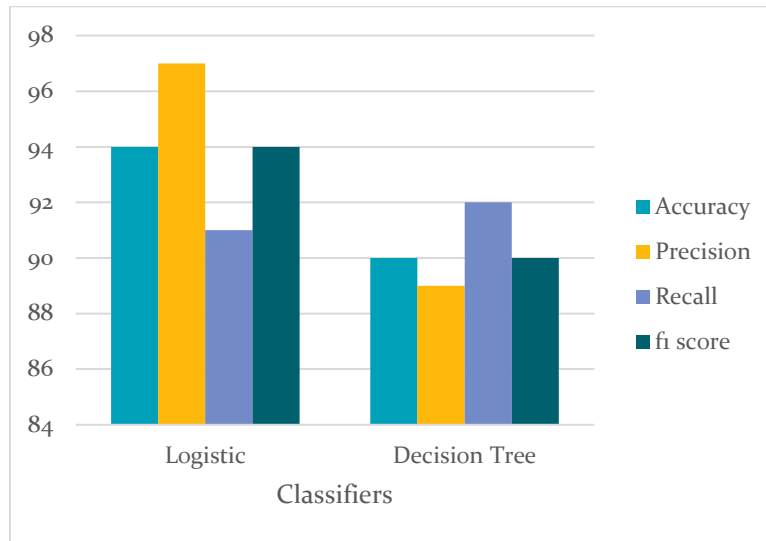
The formula for accuracy is:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

The ROC curve offers a nuanced perspective on our model's performance, particularly in binary classification scenarios such as fraud detection. By graphically plotting the True Positive Rate (sensitivity) against the False Positive Rate (1 - specificity) at various discrimination thresholds, the ROC curve illustrates how well our model distinguishes between fraudulent and legitimate transactions across a spectrum of decision boundaries.

Together, accuracy and the ROC curve analysis offer a comprehensive evaluation of our classification model's effectiveness in discerning between fraudulent and legitimate credit card transactions within our dataset.

COMPARITIVE STUDY



- Accuracy of Logistic Regression is more than Decision Tree classifier.
- Precision is more for Logistic regression.
- Recall of Decision Tree classifier more than that of Logistic regression.
- F1-score of Logistic Regression is more.

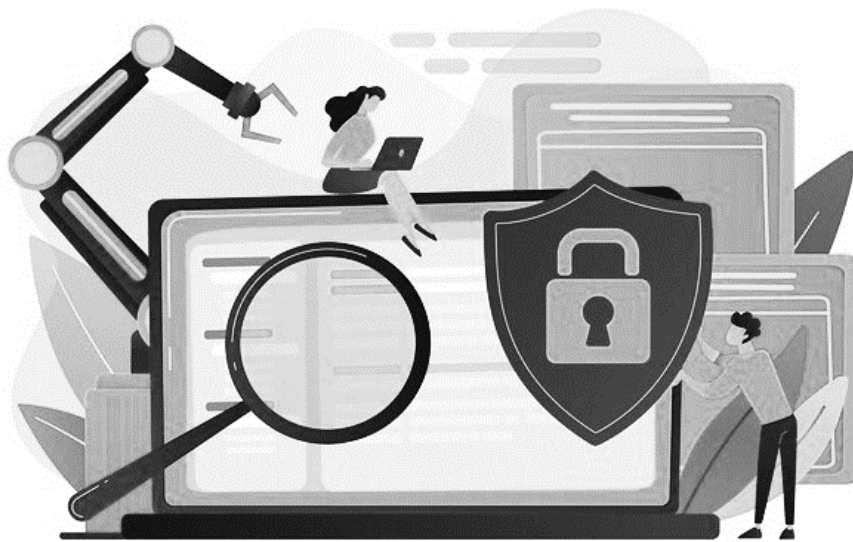
CONCLUSION

In conclusion, selecting the right classifier for ‘credit card fraud detection’ is a critical decision that hinges on multiple factors, including the nature of the data, the problem's complexity, and the desired trade-offs between various performance metrics. Our analysis is based on machine learning algorithms, like Logistic Regression and Decision Tree. Each of these classifiers comes with its own strengths and weaknesses.

Logistic Regression, with its simplicity and interpretability, may be a suitable choice when the focus is on understanding the factors influencing fraud. Decision trees can offer interpretability and handle feature importance analysis, making them suitable for cases where understanding the reasoning behind fraud detection is crucial.

Ultimately, the choice of classifier should be guided by a thorough understanding of the specific fraud detection problem, the available data, and the desired balance on the performance metrics like accuracy, precision, recall, and the F1-score.

With Respect to the above analysis, we conclude that: On the basis of accuracy, we can say that Logistic Regression gives the best results.



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